Communication Network Design
Lab session #2

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Outline for today's class

- Recap from Lab #1
  - Exercise on '*Route Formulation (RF) vs. Flow Formulation (FF)*'
  - Discussion on Task #1
- Route Formulation with Wavelength Continuity
  - Task #2: RF with Wavelength Continuity vs. RF without Wavelength Continuity'
- Elastic Optical Networks
  - Routing and Spectrum Assignment (RSA) Problem
- Heuristic Approaches
- Exercise on 1+1 protection

Recap of LAB 1
Flow Formulation (FF) vs. Route Formulation (RF)

- *'SplittableFlowFormulation.java'* (compiled and got '*.class*' file)
- *'RouteFormulationVWP.java'* (compile it to get '*.class*' file)
  - Assume k (number of shortest paths calculated a-priori) = {1, 2, 3}

- Load 'example7nodes_withTraffic.n2p' (set C = 25, 30, 35)
- Compare FF and RF in terms of capacity consumption and execution time

Formulation	k	C = 25	C = 30	C = 35	Time (sec.)
FF	NA	196	196	196	
RF	1	NA	NA	196	
	2	196	196	196	
	3				

Recap of LAB 1
Flow Formulation (FF) vs. Route Formulation (RF)

- Extension of the exercise

	k	(4) C = 16	(2) C = 17	(3) C = 18	C = 25	C = 30	(1) C = 31	C = 35	Time (s)
FF	NA	X	204	200	196				
RF	1		X	X	X	X	196	196	
	2		X	200	196				
	3	X	204	200	196				

- (1)-(4): follow these additional steps to find the values of C for which there is no solution when k = 1, 2, 3 in RF
  1) What value of C ensures a solution in case of k = 1
  2) What value of C leads to having no solution for k = 2
  3) Objective function values for C values between (2) and C = 25
  4) is there a value of C for which there is no solution also in case k = 3

Recap of LAB 1

Task 1: Splittable FF vs. Unsplittable FF

- Modify *FlowFormulation.java* to assume unsplittable flows
  - What are the differences with respect to splittable FF?

- Load the 7-nodes network topology with traffic (the network topology used during the demo)
  - Compare '*splittable*' FF & '*unsplittable*' FF varying link capacity C = {10, 15, 17, 20, 30, 40, 50}
  - Plot/tabulate total carried traffic with respect to C
  - Please refer to the task template submission form

- Submit the pdf file and code via e-mail with subject "*LAB_CND_task_#*" to qiaolun.zhang@polimi.it and memedhe.ibrahimi@polimi.it

Recap of LAB 1

Task 1: hints and task completion

1. Include "*HashSet*" and "*Set*" java libraries
   - "import java.util.HashSet;" and "import java.util.Set;"
2. An important modification to the code for unsplittable FF
   - Comment line-64 and uncomment line-65 and line-66
3. Modify objective function
4. Modify solenoidality and capacity constraints

Link Capacity (Gbps)	*Splittable*	*Unsplittable*
10		
15		
17		
20		
30		
40		
50		

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  - Discussion on Task #1
- **Route Formulation with Wavelength Continuity**
  - **Task #2: RF with Wavelength Continuity vs. RF without Wavelength Continuity'**
- Elastic Optical Networks
  - Routing and Spectrum Assignment (RSA) Problem
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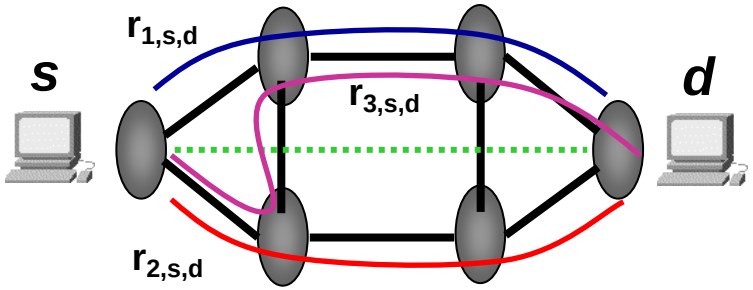
- Exercise on 1+1 protection

Route Formulation
Without wavelength continuity

- (All) possible paths between each (s, d) pair are evaluated a priori
  - Constrained routing (k shortest paths)
- Variables represent which path is used for a given connection
  - r_{cn}: path n is used by r_{cn} connections between s_c and d_c

G(N,E)	N set of Nodes, E set of Links
c	node pair (s_c and d_c)
l,k	link identifiers (source-destination pairs)
n	path identifier
V_c	number of connections requests having s_c as source and d_c as destination
$r_{c,n}$	(integer) number of connections routed on n-th admissible path for node-couple c
W	Link capacity, number of wavelengths
P_n	Number of links on path n



Route Formulation
Without wavelength continuity

- **Objective function**: minimize the total amount of capacity used (capacity consumption) to route demands on all links

$$\min \sum_{c,n} r_{cn} \cdot P_n$$

Wavelength is not accounted for: **no wavelength continuity**

Solenoidality	$\sum_{n} r_{c,n} = v_c \quad \forall c$
Capacity	$\sum_{r_{c,n} \in R_{l,k}} r_{c,n} \leq W \cdot F_{l,k} \quad \forall (l,k)$
Integrity	$r_{c,n}$ integer $\quad \forall (c,n)$ $F_{l,k}$ integer $\quad \forall (l,k)$

$R_{(l,k)}$: set of (all) admissible paths passing through link *(l,k)*

Route Formulation
With wavelength continuity

- $r_{c,n,\lambda}$ = number of connections routed on the n-th admissible path for c on wavelength λ

Solenoidality	$\sum_n r_{c,n,\lambda} = v_{c,\lambda}$	$\forall\, c,\lambda$
	$\sum_\lambda v_{c,\lambda} = v_c$	$\forall\, c$
Capacity	$\sum_{r_{c,n} \in R_{l,k}} r_{c,n,\lambda} \le F_{l,k}$	$\forall (l,k),\lambda$
Integrity	$r_{c,n,\lambda}$ integer	$\forall\, (c,n),\lambda$
	$F_{l,k}$ integer	$\forall (l,k)$
	$v_{c,\lambda}$ integer	$\forall c,\lambda$

Example with v_c (v_{sd}) = 3

Needs to be = 3

v_{c1} = 1, v_{c2} = 1, v_{c3} = 1 and all other $v_{c\lambda}$ = 0

OR

v_{c1} = 3 and all other $v_{c\lambda}$ = 0

r_{c11} = 1, r_{c12} = 1 and r_{c13} = 1

or r_{c21} = 1, r_{c32} = 1 and r_{c13} = 1

OR

r_{c11} = 3 or r_{c21} = 3 or ..

Task #2
RF with and without wavelength continuity

- '*RouteFormulationWP.java*' contains some MISSING lines of code
  - Add these lines to assume wavelength continuity!
- Load the network topology from WeBeep
  - All task-related material can be found in "*02-LAB-CND-2*" folder, sub-folder "*03 Task #2*"
    - Network topology file "*example7nodes_withTraffic_task2.n2p*"
- Compare RF with and without wavelength continuity varying link capacity C = {20, 23, 25, 27, 30}, considering k = 1, 2, 3
  - Tabulate total capacity consumption on all links with respect to the capacity C
- Share pdf and code via e-mail with the subject "LAB_CND_task_2" and name the file with your codice persona
- **DEADLINE: 08/11/2024 (before CND class)**

Task #2
Hints to complete the task

1. Ensure you define the objective function correctly
   - You need to ensure parameter P_n (a vector) multiplies variable $r_{cn\lambda}$ (You must use a function of the type "sum (matrix, dimension)"
   - http://www.net2plan.com/jom/gettingStarted.php

$$\min \sum_{c,n,\lambda} r_{cn\lambda} \cdot P_n$$

2. In the capacity constraint: make sure that every wavelength can be used only once (you have only one fiber connecting two nodes)
   - You can set the number of wavelengths according to "linkcapacity" or similar to how you set "k" in line 31
3. Make sure you are considering decision variables to be integer
4. Ensure the value of objective function is returned
   - Comment this line: "*//return "Ok! Total number of wavelengths used in the links: " + netPlan.getVectorLinkCarriedTraffic().zSum();*"
   - Add this line instead: "*return "Ok! Total number of wavelengths used in the links: " + op.getOptimalCost();*"

C	WP with continuity	vWP without Continuity
20		
23		
…		
30		

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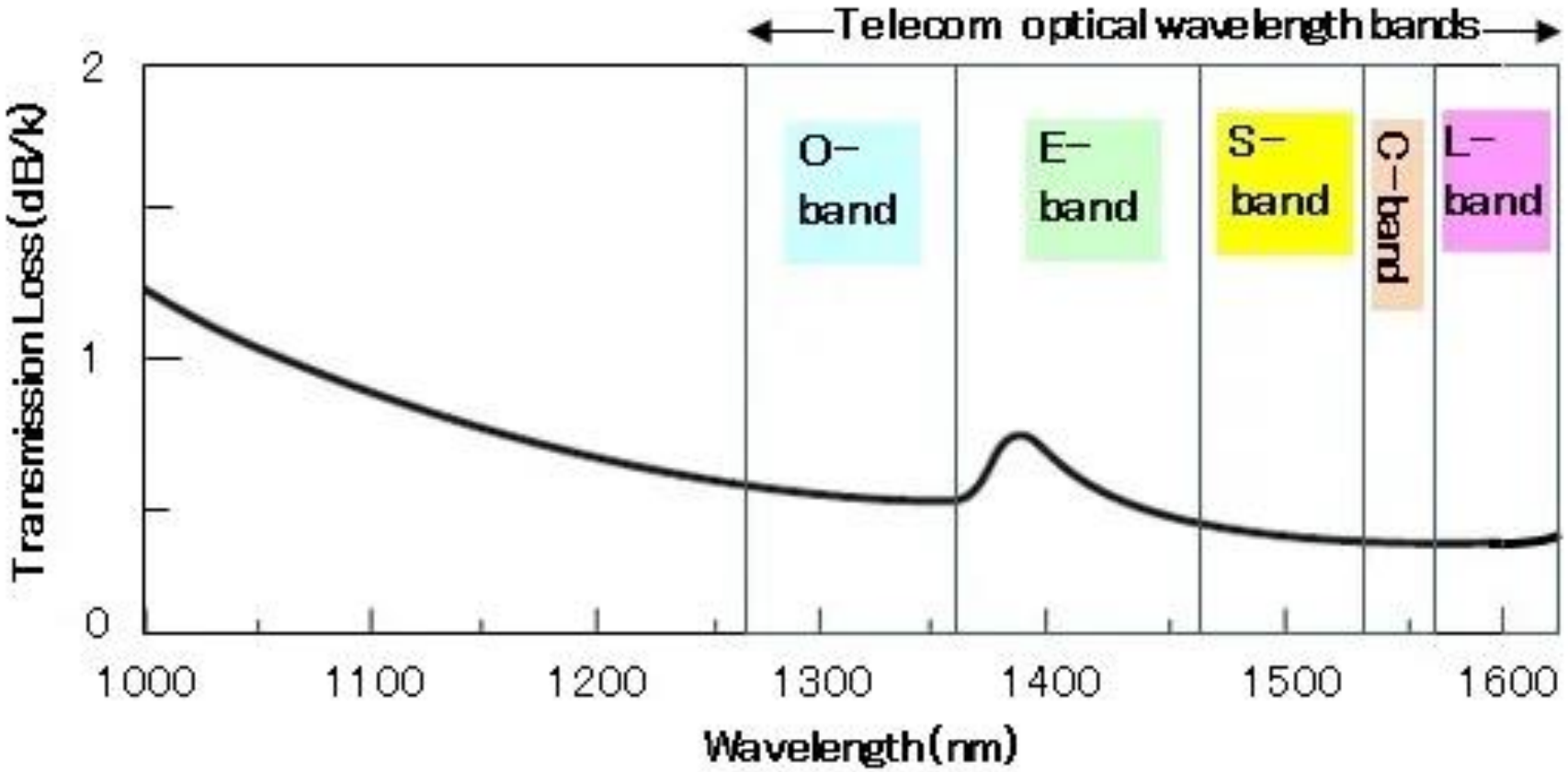
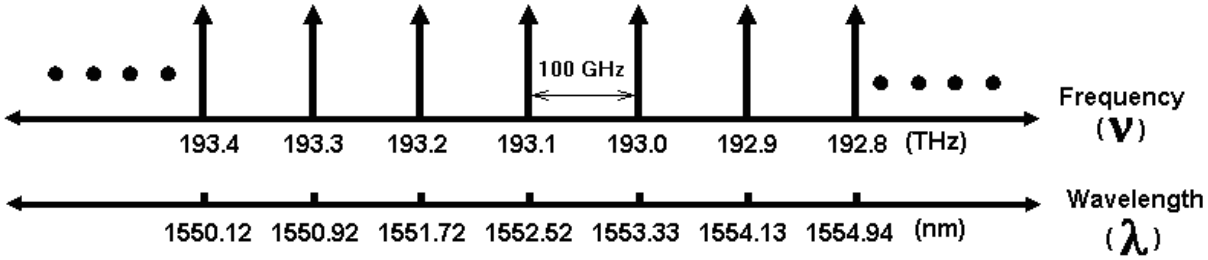
- **Elastic Optical Networks**
  - **Routing and Spectrum Assignment (RSA) Problem**
- **Heuristic Approaches**
- Exercise on 1+1 protection

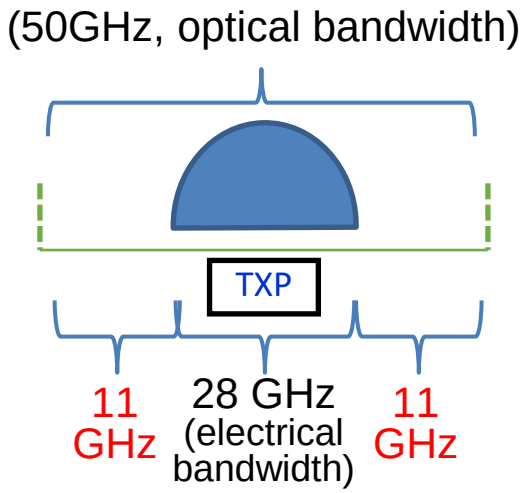
Elastic Optical Networks
Fixed grid vs Flexible grid



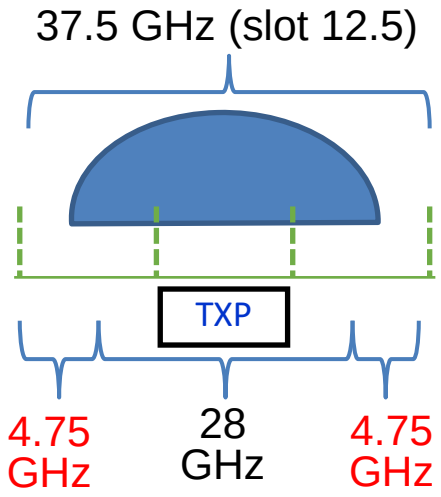
Elastic Optical Networks
Fixed grid vs Flexible grid

- Current Wavelength Division Multiplexing (WDM)
  - Fixed channel spacing (50/100 GHz)



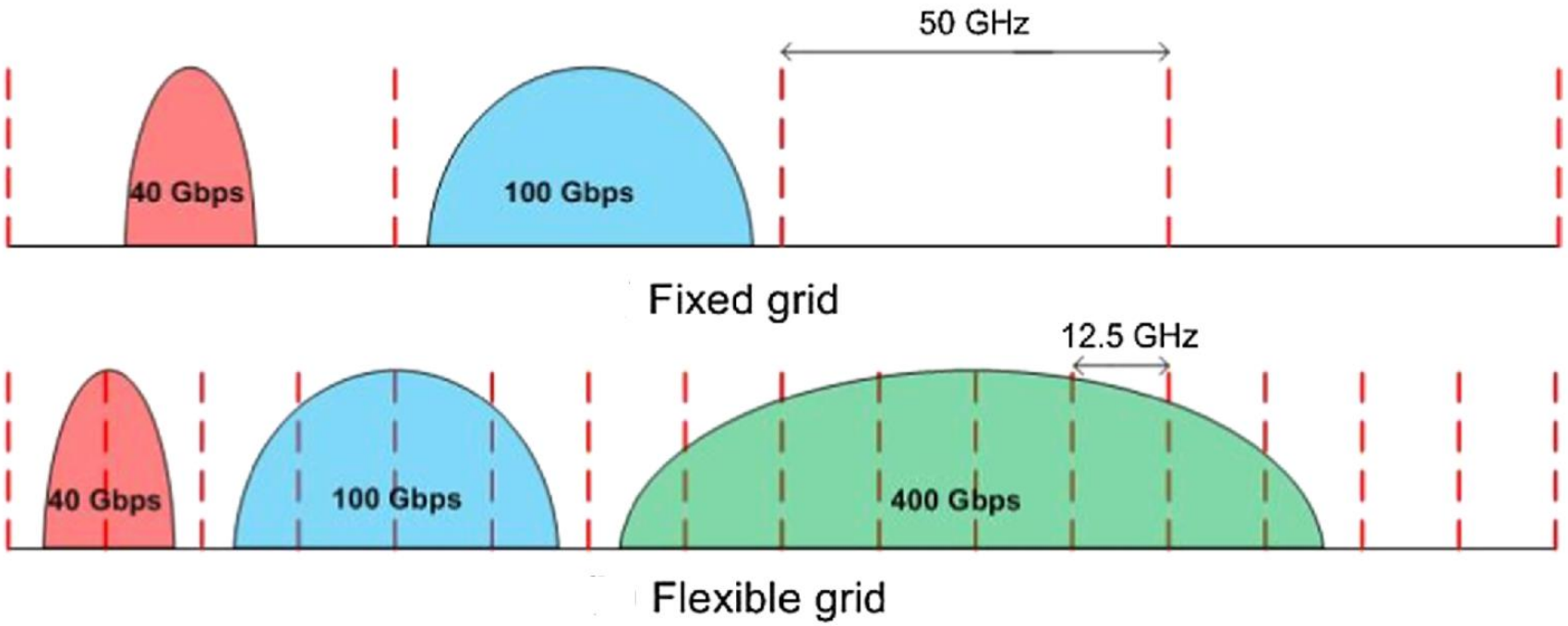


- Flexible grid (Flexi Grid): frequency slices with finer granularity (12.5 GHz)
  Increased efficiency in the usage of spectral resources
  - 100 Gb/s = 28 GHz, 28 Gbaud
  - 28 Gbaud are divided into:
    (1-25) Gbaud is the data transmission rate
    (2-3) Gbaud for FEC (forward error correction)
  - with DP-QPSK: 25 GHz x 4 bit/Hz = 100 Gb/s



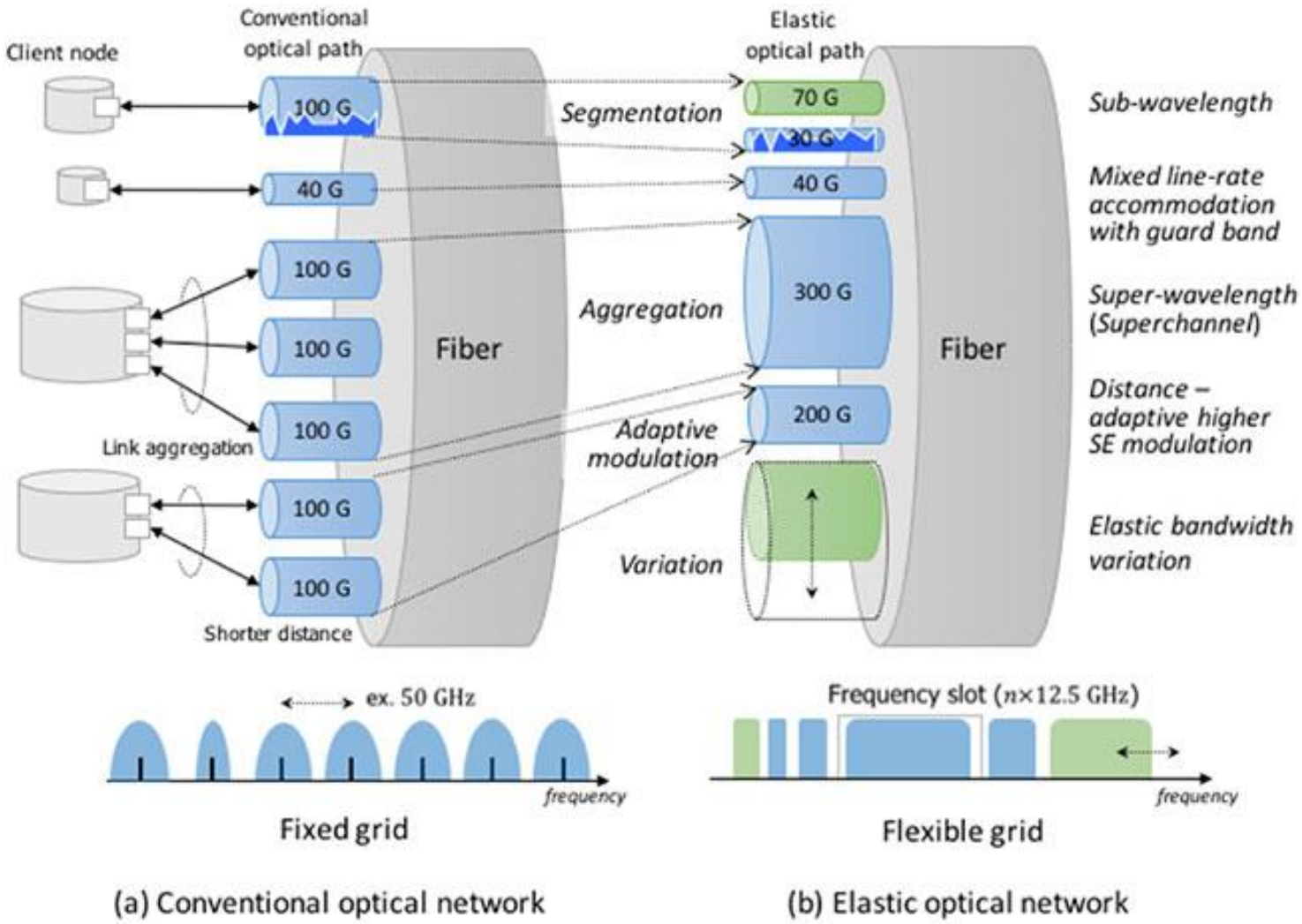
Elastic Optical Networks
Fixed grid vs Flexible grid

- Fixed Grid: fixed channel spacing (50 GHz)
- Flexible grid (Flexi Grid): frequency slices with finer granularity (12.5 GHz)

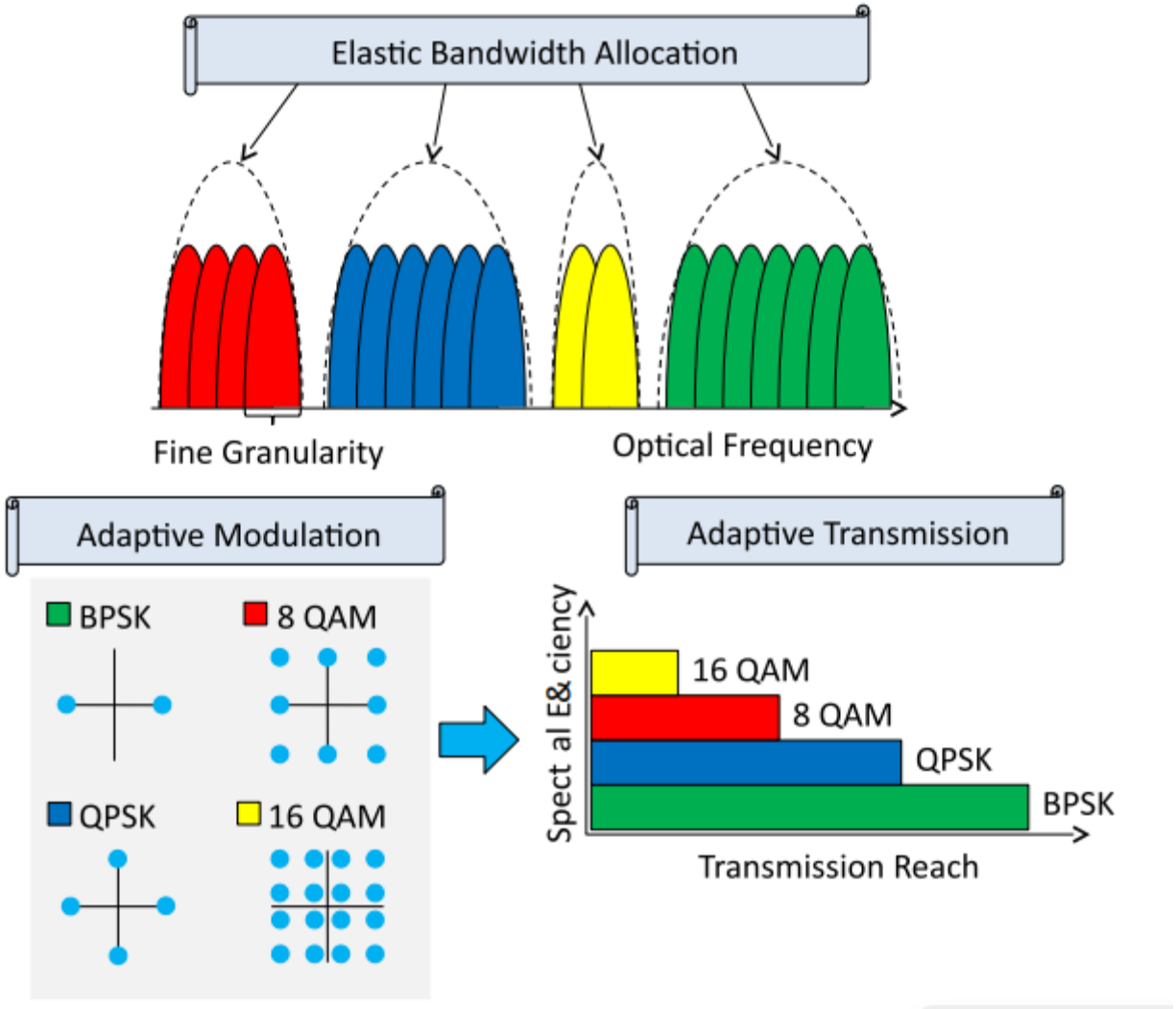


Routing and Spectrum Assignment (RSA) instead of traditional Routing and Wavelength Assignment (RWA)

Elastic Optical Networks
Fixed grid vs Flexible grid

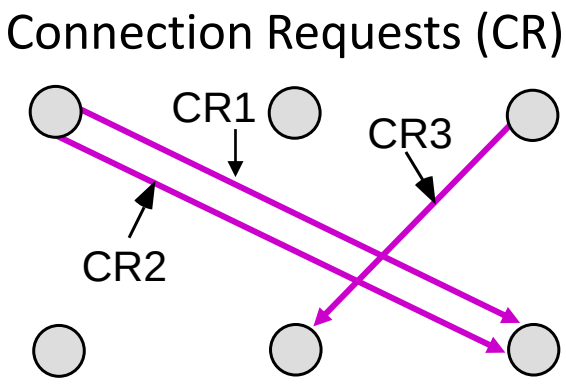


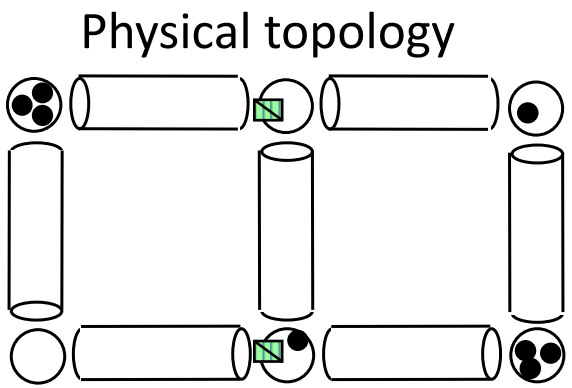
Elastic Optical Networks
Fixed grid vs Flexible grid

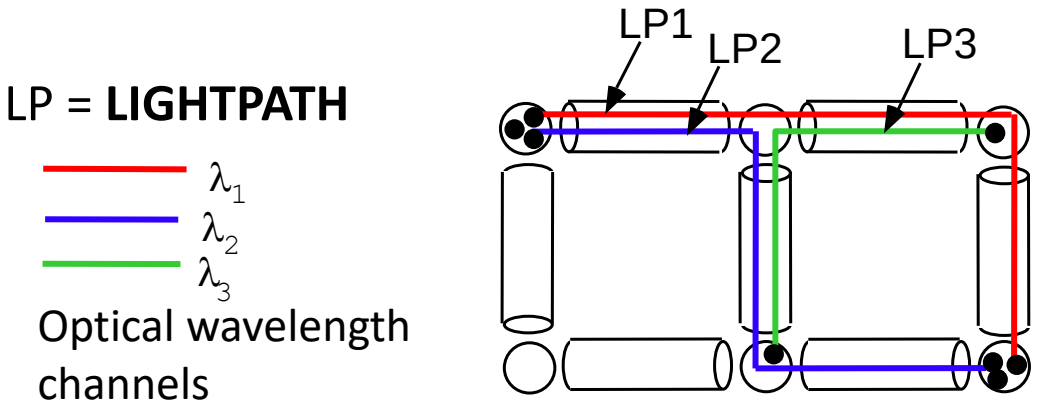


Elastic Optical Networks
RWA vs. RSA

- In RWA, CRs require a wavelength (e.g., 50 GHz spectrum)
  - Wavelength continuity constraint
- In RSA, CRs require frequency slots that are multiple of Frequency Slice Units (FSUs), e.g., 12.5 GHz
  - Spectral continuity constraint
  - Spectral contiguity constraint



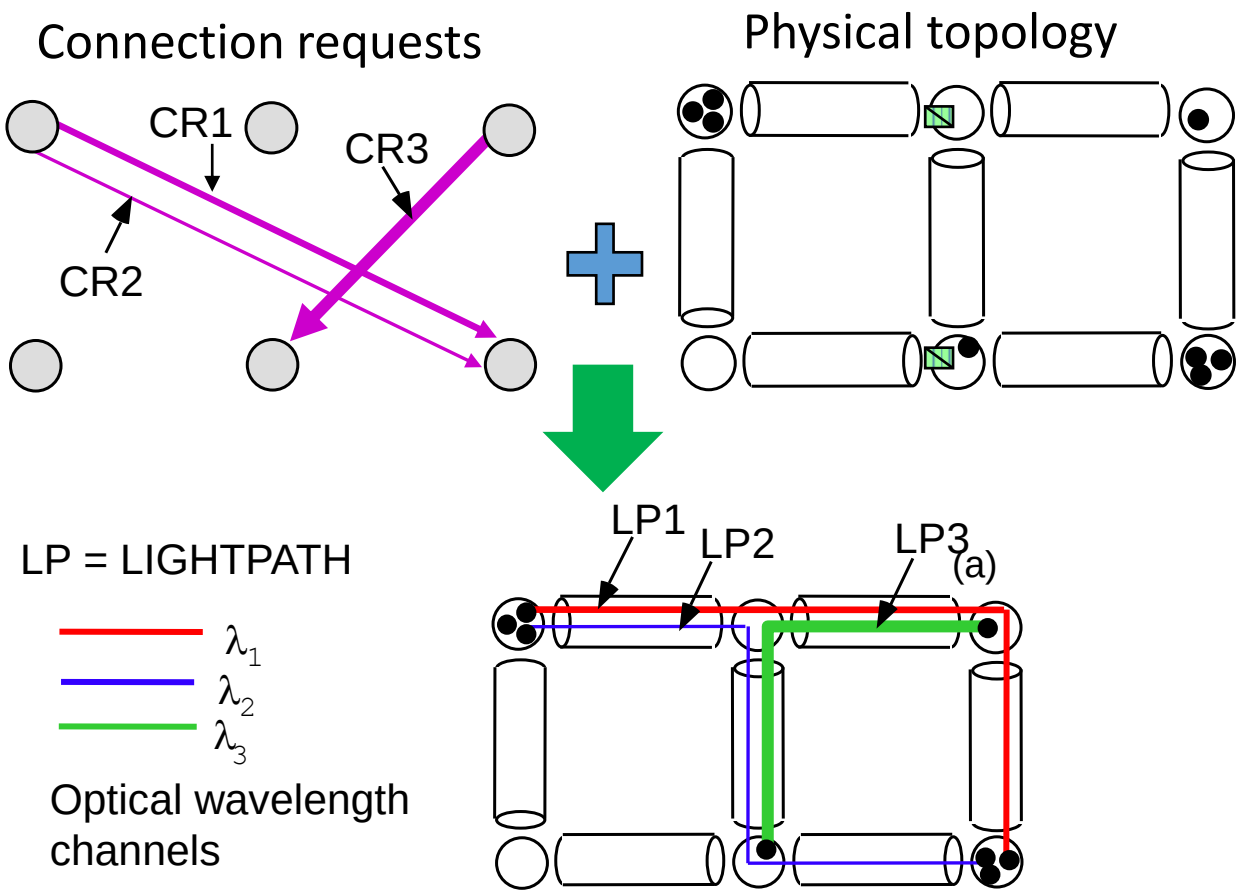




Elastic Optical Networks
RWA vs. RSA

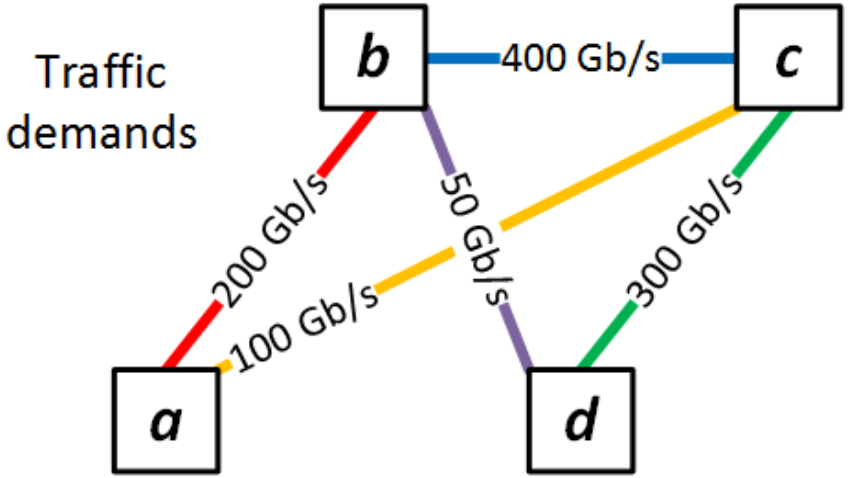
- CRs require frequency slots that are multiple of the basic frequency slice

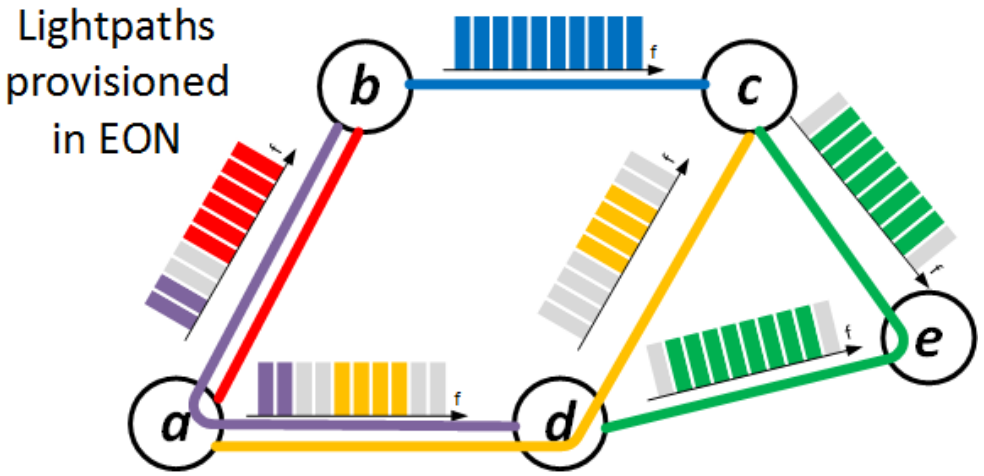
- Main constraints to solve RSA
  - Route CRs
  - Spectrum continuity
  - Spectral adjacency (aka “spectrum contiguity”)
  - Link capacity



Elastic Optical Networks
Continuity and contiguity constraints

- **Continuity constraint**: demand must use exactly the same spectrum slots in all links over which it is routed (e.g., green and purple spectrum)
- **Adjacency (contiguity) constraint**: slices assigned to a particular demand must be adjacent (contiguous)





Heuristic Approaches
Heuristic vs. ILP

- Heuristic approaches
  - Practical methods, not guaranteed to be optimal but are good enough
  - An alternative to Mathematical Programming for realistic dimension problems

- Optimization approaches
  1. **Greedy**
  2. Local Search
  3. Meta-heuristics (e.g., Genetic Algorithms)

(Example) Comparing performances of an ILP and a heuristic for RSA in two networks considering full mesh traffic matrix
Objective function: Minimize overall spectrum consumption

7-node network	ILP	Heuristic
Value of Obj. ft.	200	205
Execution time (sec)	150	2

52-node network	ILP	Heuristic
Value of Obj. ft.	NA	8000
Execution time (sec)	NA	25

Heuristic Approaches
Greedy heuristics

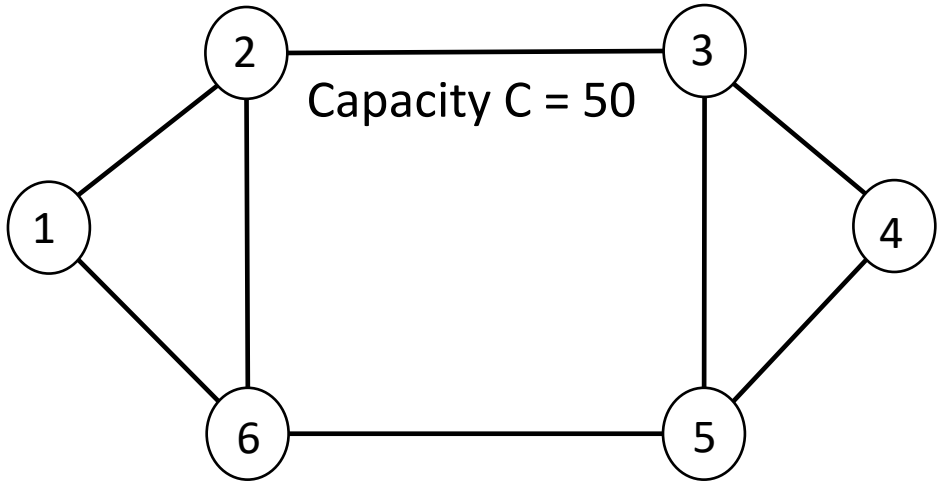
- Constructive algorithms that start with an empty solution. A part of the solution is constructed in each iteration
  - Based on the optimization of a cost function
  - No coming back is allowed: parts of solution already set do not changes

- Stopping condition: When does it stop?
  - Maximum running time/iterations
    - After some fixed time or a maximum number of iterations
  - Maximum time/iterations without improvement
    - If no improvement in the best solution found so far is observed in a given amount of time or number of iterations
  - Approximating an optimality bound
    - When the solution is near to the solution obtained with an approximation technique

Heuristic Approaches
Greedy heuristic for the RSA problem

- **Given** network topology and set of traffic demands each requesting a number of Frequency Slot Units (FSUs)
- **Decide** route and spectrum assignment for each demand
- **Objective**: minimize overall amount of carried traffic on all links
- **Subject to** (while respecting) RSA constraints (link capacity, continuity, etc..)

Network Topology



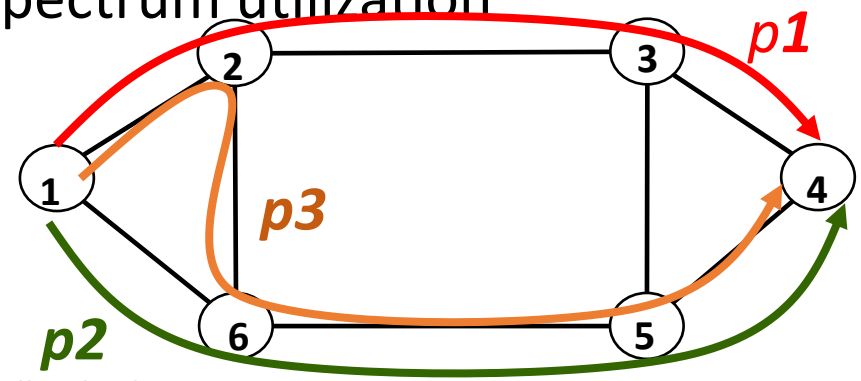
Traffic Demands

Demand id	*Src*	*Dst*	*#FSUs*
D1	1	4	3
D2	2	4	3
D3	5	1	3
D4	6	2	3
D5	3	5	6

Heuristic Approaches
Greedy heuristic for the RSA problem

1. Sort demands (*in which order?)*
2. Compute candidate paths for each demand (*how many?)*
3. For each demand
   a) Sort paths in increasing order of number of links.
   b) Consider shortest path: If enough capacity, route and allocate spectrum. Otherwise, consider next path in list.
   c) Update number of FSUs available on each link used
4. Consider next demand in list, stop when no more demands need to be routed
5. Output RSA of each demand and overall spectrum utilization

Demand id	*Src*	*Dst*
D1	1	4



Heuristic Approaches

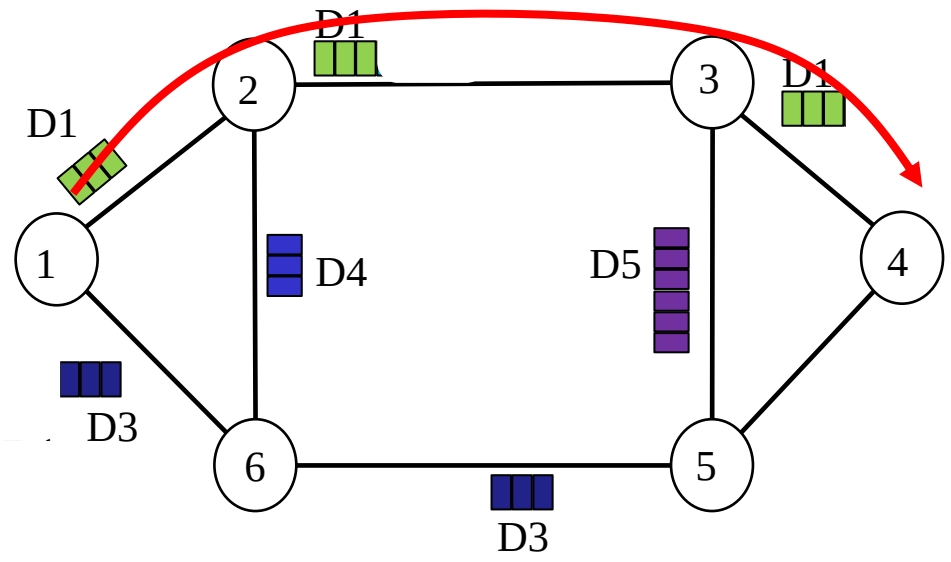
Greedy heuristic for the RSA problem

- For each demand
  - Find best path, route and allocate spectrum
  - Update number of FSUs on each fiber

id	Src	Dst	#FSUs
D1	1	4	3
D2	2	4	3
D3	5	1	3
D4	6	2	3
D5	3	5	6

Shortest path available? If yes then route, else check 2nd shortest path

Link id	Available Capacity	Utilized Capacity
1,2	~~50~~ 47	3
2,3	~~50~~ ~~47~~ 44	6
3,4	~~50~~ ~~47~~ 44	6
5,6	~~50~~ 47	3
6,1	~~50~~ 47	3
6,2	~~50~~ 47	3
3,5	~~50~~ 44	6



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- Heuristic Approaches
- **Exercise on 1+1 protection**

Exercise on '1+1 protection'

- Load «*Example7nodes.n2p*»
- Load the traffic matrix «*tm7nodes.n2p*»

- Load and run the heuristic (the '.java' file from WeBeep) *'Offline_ipOverWdm_routingSpectrumAndModulationAssignmentHeuristicNotGrooming.java'*
- Run algorithm once so that IP layer is shown in '*View/Edit network state*'
- Set offered traffic per demand in **IP layer** tab to 10 Gbps

- Set numFrequencySlotsPerFiber (capacity) = 40
  - Spectrum resources in a fiber to serve the traffic requests
- Other parameters as defined by defaults

Exercise on '1+1 protection'

- Run/Execute the heuristic approach with '*no protection*' and then with '*1+1protection*'
- Under algorithm settings, choose the following
  - No protection: NetworkRecoveryType: ***not-fault-tolerant***
  - 1+1 Protection: NetworkRecoveryType: ***1+1 srg-disjoint-lps***
- Repeat the same for offered traffic per demand = 20 Gbps, 30 Gbps and 40 Gbps
  - What is the value of the objective function?

Objective function for varying traffic and varying numFrequencySlotsPerFiber

Traffic (Gbps)	not-fault-tolerant (no protection)	1+1 srg-disjoint-lps (1+1 protection)
10		
20		
30		
40		
80		

Exercise on '1+1 protection'

- What is the lowest number of frequency slots per fiber which ensures all demands are protected with 60Gbps (1+1 protection)?

numFreqSlots	***not-fault-tolerant*** *(no protection)*	***1+1 srg-disjoint-lps*** *(1+1 protection)*
40		
50		
60		
70		
80		

- What if we change value of k? (k=3 by default)
  - Does a higher k (e.g., k = 10) ensure a solution for 60Gbps (1+1 protection) with a lower number of allocated frequency slots?

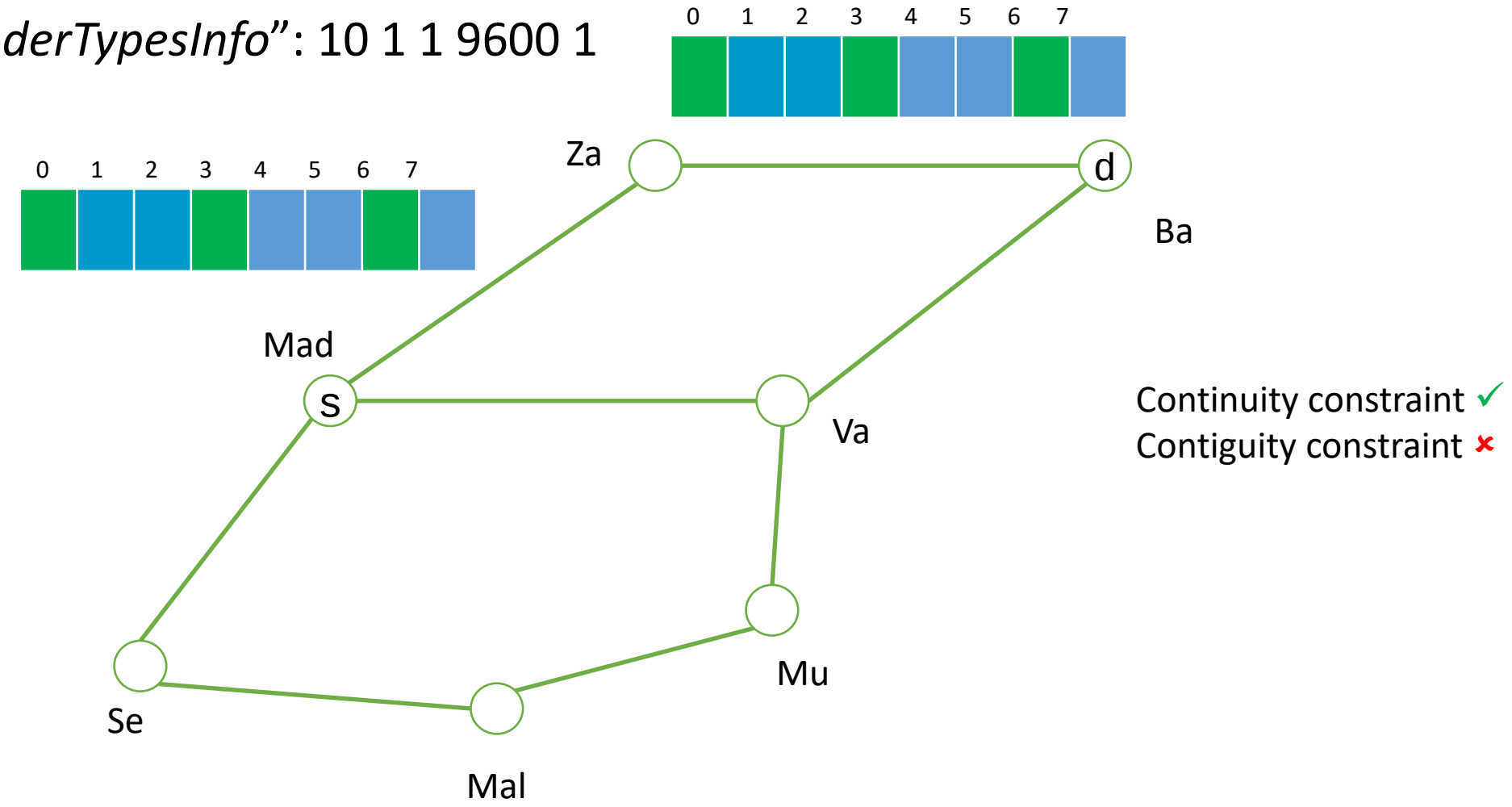
Changing transponder capacity and spectrum occupancy

- "*transponderTypesInfo*": **10** 1 **1** 9600 1
  - Default: 10 1 1 9600 1 Transpoder types separated by ";" .
    Each type is characterized by the space-separated values: (i) Line rate in Gbps, (ii) cost of the transponder, (iii) number of slots occupied in each traversed fiber, (iv) optical reach in km (a non-positive number means no reach limit), (v) cost of the optical signal regenerator (regenerators do NOT make wavelength conversion ; if negative, regeneration is not possible).

- Set offered traffic to 30 Gbps; numFreqSlots = 40
  a) If number of slots for each transponder is one (default) – can we ensure the contiguity constraint?
     - E.g., "*transponderTypesInfo*": 10 1 1 9600 1
  b) What if we modify the number of slots for each transponder – is there any difference compared to previous case?
     - E.g., "*transponderTypesInfo*": 30 1 3 9600 1

Changing transponder capacity and spectrum occupancy

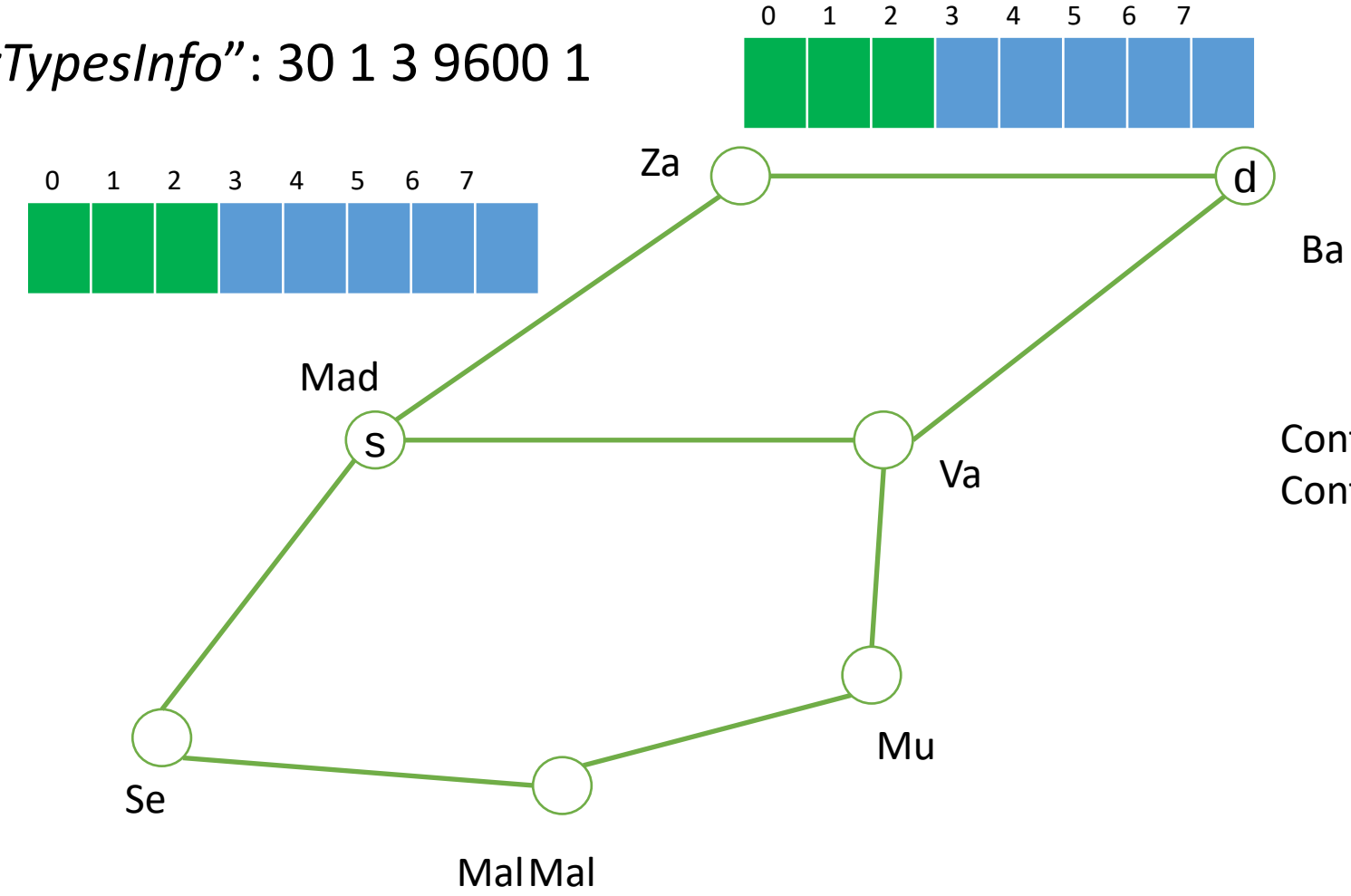
"*transponderTypesInfo*": 10 1 1 9600 1



Continuity constraint ✓
Contiguity constraint ✗

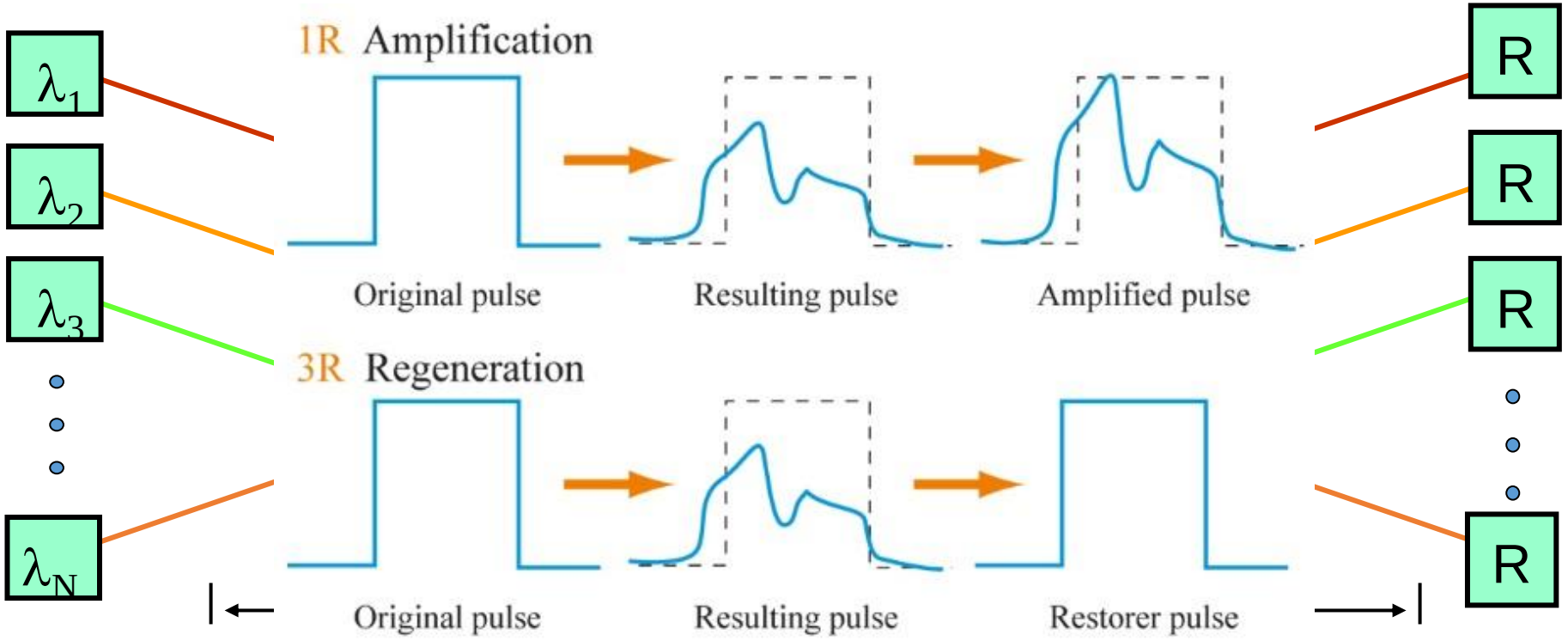
Changing transponder capacity and spectrum occupancy

"*transponderTypesInfo*": 30 1 3 9600 1



Continuity constraint ✓
Contiguity constraint ✓

Amplification and 3R regeneration

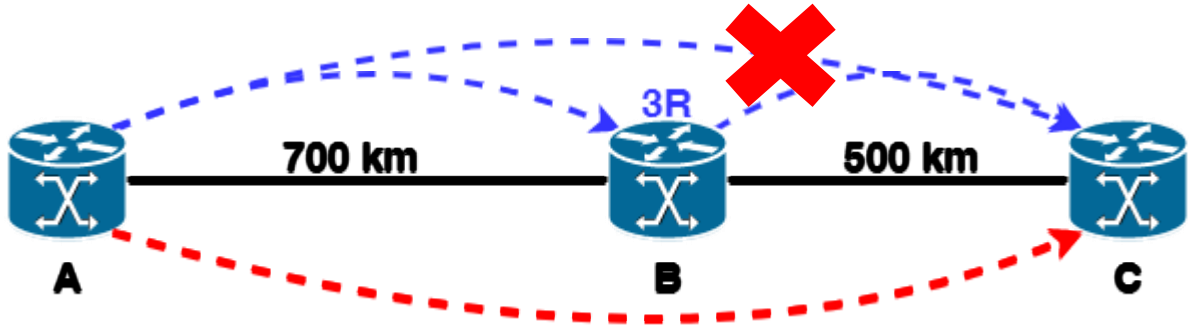


- In general signal regeneration can be 1R, 2R or 3R
  - Pulse re-amplification, re-shaping, re-timing
  - Optical amplification is 1R
- When should the signal be re-generated? How can we know?!

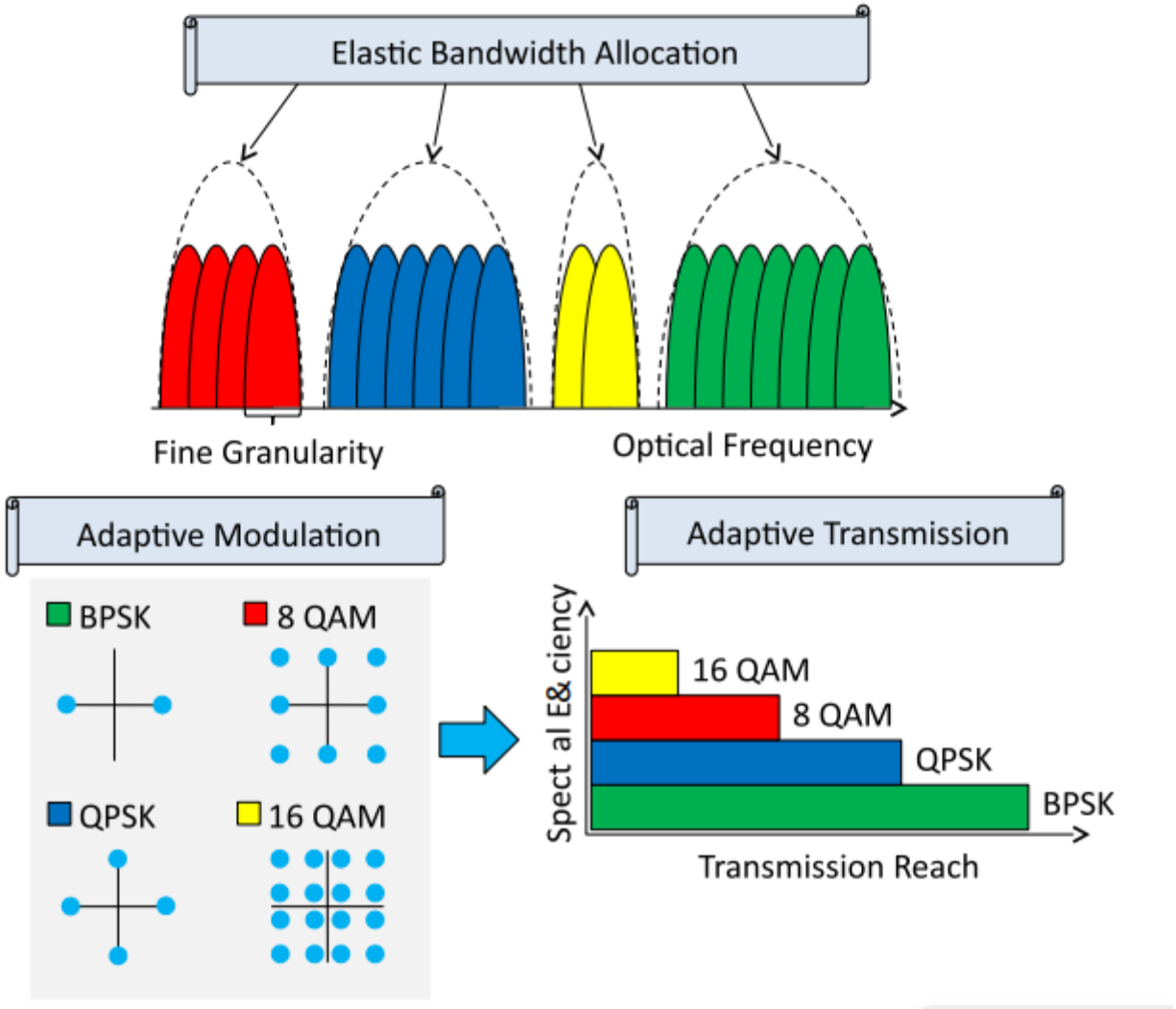
Amplification and 3R regeneration

Data Rate (Gb/s)	Modulation format	Bits/symbol (Gb/s) - Entropy	Channel spacing Δf (GHz)	Reach (km)
800	PCS 64 QAM	5.67	100	150
700	PCS 64 QAM	5.00	100	400
600	16 QAM	4.00	100	700
500	PCS 16 QAM	3.60	100	1300
400	PCS 16 QAM	3.00	100	2500
300	PCS 16 QAM	2.39	100	4700
300	64 QAM	6.00	50	100
200	16 QAM	4.00	50	900
100	QPSK	2.00	50	3000

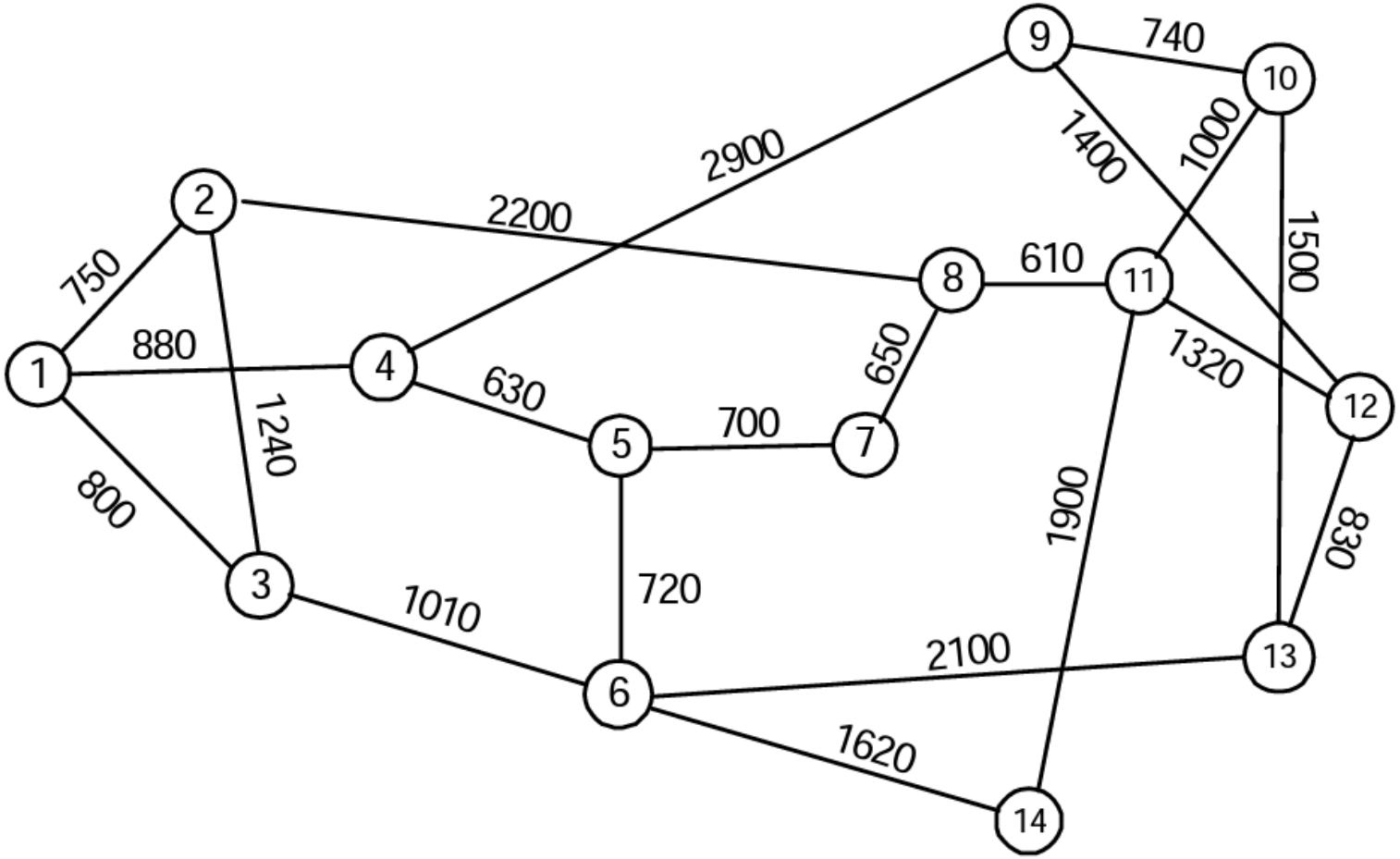
CR1: (A, C) 100 Gb/s, QPSK CR2: (A, C) 200 Gb/s, 16-QAM



Elastic Optical Networks
Fixed grid vs Flexible grid



Amplification and 3R regeneration



- Route traffic between each-node pair and minimize number of 3R regenerators